

1. Administrative & Study Identification

Full Study Title: A Phase 1 Study of SEA-CD70 in Myeloid Malignancies **Short Title/Acronym:** SEA-CD70 in MDS and AML
Protocol Number: SGNS70-101 (C5781001) **Posting ID:** 744516658770387832 **NCT Number:** NCT04227847 **Sponsor/Company Name:** Pfizer (Seagen, a wholly owned subsidiary of Pfizer)
Investigational Product: SEA-CD70 (PF-08046040), a nonfucosylated CD70-directed monoclonal antibody **Phase of Development:** Phase 1
Medical Monitor: Pfizer Clinical Trials Medical Monitor

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the safety, tolerability, and to identify the maximum tolerated dose (MTD) or recommended expansion dose of SEA-CD70 monotherapy and in combination with azacitidine (with or without venetoclax) in participants with myeloid malignancies. **Secondary Objectives:** To evaluate preliminary anti-tumor activity, including Objective Response Rate (ORR), Complete Remission (CR) rate, time to response (TTR), duration of response (DOR), and to characterize the pharmacokinetics (PK) of SEA-CD70.

2.2 Background & Rationale To address the critical unmet medical need in patients with relapsed/refractory or higher-risk Myelodysplastic Syndromes (MDS) and Acute Myeloid Leukemia (AML) / Leukemia. CD70 is aberrantly expressed on myeloid leukemic blasts and drives proliferation through CD70-CD27 signaling. SEA-CD70 is engineered with enhanced effector functions to eliminate CD70+ leukemic cells via antibody-dependent cellular cytotoxicity/phagocytosis (ADCC/ADCP) and to block oncogenic CD27 signaling, presenting a promising targeted therapeutic approach.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional (Dose-escalation and Dose-expansion) **Control Type:** Single-arm (Multiple cohorts: Parts A through G) **Blinding:** Open-label **Randomization:** N/A

3.2 Planned Enrollment & Duration Targeted Enrollment: 178 evaluable patients (across 7 cohorts/parts) **Trial Status:** RECRUITING (Currently recruiting participants) **Number of Sites:** Multicenter (Global, including comprehensive cancer centers in the US and EU) **Estimated Study Start:** Early 2020

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Age ≥ 18 years (Specifically restricted to 18-75 years for Part C). 2. Eastern Cooperative Oncology Group (ECOG) Performance Status of 0-2 (Restricted to 0-1 for Part A). 3. Cytologically/histologically confirmed MDS or MDS/AML (WHO 2016/ICC 2022 criteria), either relapsed/refractory (e.g., HMA-failure) or previously untreated higher-risk (per IPSS-M). 4. Diagnosis of AML, either previously untreated and ineligible for standard induction chemotherapy, or relapsed/refractory to prior regimens (e.g., received 2-3 previous regimens, or 1 previous regimen if primary resistant, ≥ 60 and ≤ 75 years old, first CR duration < 6 months, or adverse-risk).

4.2 Exclusion Criteria 1. Previous exposure to CD70-targeted therapeutic agents. 2. Prior allogeneic hematopoietic stem cell transplant for any condition. 3. Central nervous system (CNS) leukemia. 4. History of clinically significant sickle cell anemia, autoimmune hemolytic anemia, or idiopathic thrombocytopenic purpura (ITP). 5. Prior oral HMA or oral HMA-combinations (applicable to Parts D, F, and G). 6. Conditions precluding enteral administration, concomitant use of strong/moderate CYP3A inducers, or history of myeloproliferative neoplasm (applicable to Part G).

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: SEA-CD70 administered via intravenous (IV) infusion (dose-finding evaluating varying ranges), according to cohort assignment. **Route of Administration:** IV for SEA-CD70. Combined with azacitidine (Trade names: Vidaza, Onureg) in specific cohorts (IV/SC), and venetoclax (Oral) in Part G. **Duration of Treatment:** Administered in 28-day cycles until documented disease progression, unacceptable toxicity, or investigator decision.

5.2 Concomitant & Prohibited Therapy Permitted: Standard supportive care for cytopenias, including anti-infective prophylaxis and transfusion support. **Prohibited:** Strong or moderate CYP3A inducers (specifically in the venetoclax combination cohort, Part G), and prior oral HMA/HMA-combinations in specific parts (Parts D, F, and G).

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, Medical History, Baseline Bone Marrow Biopsy/Aspirate, ECOG scoring
Baseline (Cycle 1)	Day 1	First Dose of SEA-CD70, PK sampling, Safety Labs (CBC, chemistry)
Interim	Every 28 Days	Safety Labs, PK/ADA collection, Efficacy Assessment (Bone

Marrow Aspirate/Biopsy every few cycles), AE tracking | | **End of Study** | +30 Days | Final Safety Follow-up, IP Accountability, Transition to Survival Follow-up |

7. Outcome Measures (Endpoints)

7.1 Primary Endpoints 1. Number of participants with adverse events (AEs). 2. Number of participants with laboratory abnormalities. 3. Number of participants with a dose-limiting toxicity (DLT) at each dose level (Evaluated for Parts A and D only).

7.2 Secondary & Exploratory Endpoints 1. Objective Response Rate (ORR), defined as best response of CR, CRh, or PR per modified IWG criteria. 2. Pharmacokinetic profile of SEA-CD70, including:
- Area under the plasma concentration-time curve (\$AUC\$) - Maximum observed plasma concentration (\$C_{\text{max}}\$) - Time to maximum concentration attained (\$T_{\text{max}}\$)

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous evaluation at every study visit, including vital signs, physical exams, and clinical laboratory testing. **Serious Adverse Events (SAEs):** Must be reported to the Sponsor (Pfizer/Seagen) within 24 hours of site personnel awareness. **Data Monitoring Committee (DMC):** An internal Safety Monitoring Committee oversees dose-escalation decisions, cohort expansions, and overall patient safety.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Safety Analysis Set (all patients who receive \$\ge 1\$ dose of study drug); Efficacy Evaluable Set (all patients who receive study drug and have \$\ge 1\$ post-baseline disease assessment). **Sample Size Justification:** Target enrollment of 178 patients. Powered descriptively for a Phase 1 trial utilizing standard dose-escalation statistical rules, with expansion cohorts comprising approximately 10-20 patients each to refine the safety profile and estimate initial efficacy signals. **Handling of Missing Data:** Patients without post-baseline disease assessments are generally treated as non-responders for the ORR endpoint.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) guidelines and the Declaration of Helsinki. **Monitoring Plan:** Routine onsite and remote monitoring

visits conducted by the Sponsor or delegated CRO to ensure protocol adherence and data integrity. **Audit Trail:** Secure Electronic Data Capture (EDC) system with full audit trails, strict eCRF locking protocols, and query resolution tracking.

11. Study Identifiers & Governance

Primary ID: SGNS70-101 (C5781001) **Registry Number:** NCT04227847
Posting ID: 744516658770387832 **Sponsor/Lead Organization:** Pfizer (Seagen, a wholly owned subsidiary of Pfizer) **Study Title:** A Safety Study of SEA-CD70 in Patients With Myeloid Malignancies
Official Regulatory Title: A Phase 1 Study of SEA-CD70 in Myeloid Malignancies **Preferred UMLS Name:** Myeloid Neoplasms

12. Clinical Framework (Myeloid Malignancy Specific)

Primary Conditions: Myelodysplastic Syndrome, Acute Myeloid Leukemia (Leukemia) **Diagnosis Threshold:** Cytological/histological confirmation per 2016 WHO / 2022 ICC guidelines with excess blasts. **Medical Methodology:** Targeted Immunotherapy (CD70-directed ADC) alone or combined with Hypomethylating Agents (HMA) such as azacitidine. **Optimization Target:** Blast clearance, hematologic recovery, and CR/CRh achievement.

13. Study Procedures & Methodology

Management Pathway: - **Part A & D:** Stepwise dose-finding/optimization for SEA-CD70 monotherapy and combination therapy, respectively. - **Part B, C, E, F:** Cohort dose-expansion targeting distinct disease statuses (MDS, AML, untreated vs. relapsed/refractory). - **Part G:** Dose-finding evaluating SEA-CD70 + azacitidine + venetoclax in previously untreated AML unfit for chemo. **Education Protocol (HCP):** Site Initiation Visits (SIV) focusing on infusion reaction management and dose-limiting toxicity (DLT) identification. **Education Protocol (Patient):** Informed consent process outlining intensive blood draw/bone marrow sampling requirements. **Quality Audit Mechanism:** Centralized vs. local pathology reviews and strict protocol deviation tracking.

14. Observation & Follow-Up Schedule

Total Duration: Until disease progression, unacceptable toxicity, or withdrawal of consent. **Onsite Visit Cadence:** Frequent visits during Cycle 1 (e.g., Days 1, 8, 15), transitioning to Day 1 of each subsequent 28-day cycle. **Remote Monitoring Frequency:** Continuous safety tele-monitoring as clinically indicated.

Checklist Review Interval: End of each treatment cycle for dose modification/continuation clearance.

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	_____	[DD-MMM-YYYY]				